

Fine-Tuning PaliGemma-3B Model Using QLoRA with Remote Sensing Dataset

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I. INTRODUCTION

The transformer architecture [1] and its vision adaptation ViT [2] established the foundation for modern Vision-Language Models (VLMs). In this project, we use **PaliGemma**, a vision-language model combining SigLIP (vision encoder) and Gemma (language decoder), for image-captioning task on a remote sensing dataset. PaliGemma is a general-purpose pre-trained VLM intended as a base model for downstream fine-tuning on domain-specific tasks. We fine-tune the baseline PaliGemma with the dataset that was provided using QLoRA PEFT, and evaluate them to see how fine-tuning the model with remote-sensing image-captioning dataset affects the model.

II. DATASET

The RISC Dataset includes 44,521 satellite images and 5 different captions for each of the image samples briefly describing the image, total of 222,605 captions. Dataset can be found in https://huggingface.co/datasets/caglarmert/full_riscm

III. METHODS

Before training, %90 of the dataset is split as a training set and %10 as a test set. The first caption of the 5 different captions is selected for the process.

We used 4-bit NF4 quantization with bf16 compute dtype using bitsandbytes package; LoRA adapters with $r=8$, $\alpha=16$, $\text{dropout}=0.05$, applied to attention projections (q-proj, k-proj, v-proj, o-proj) on the Gemma language decoder only. the SigLIP vision encoder and multi-modal projector remained frozen since the visual features generalize well across natural and remote sensing imagery. 11.3M trainable parameters were trained. (0.39% of total parameters 2.93B).

Fine-tuning was done with effective batch size 4 (per-device batch 1, gradient accumulation 4), learning rate $2e-4$ with AdamW-8bit optimizer, 800 optimization steps on shuffled training set, bf16 mixed precision. Kaggle Tesla T4 (16 GB) was used as the accelerator. Note that fine-tuned for 800 steps (approximately 8% of one epoch) due to compute constraints; this proved sufficient to demonstrate effective difference after fine-tuning.

TABLE I
BLEU SCORES ON 100 HELD-OUT TEST SAMPLES

Metric	Baseline	QLoRA Fine-Tuned
BLEU-4 (overall)	0.000	0.318
BLEU-1 precision	0.170	0.600
BLEU-2 precision	0.034	0.374
BLEU-3 precision	0.004	0.278
BLEU-4 precision	0.000	0.219
Brevity penalty	0.084	0.931
Length ratio	0.287	0.934

To evaluate the baseline and fine-tuned model, BLEU scores are computed for each model, first caption is used as the ground truth. 100 samples from the test set were used to compute the scores.

IV. RESULTS

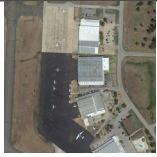
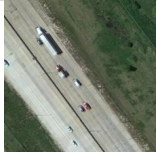
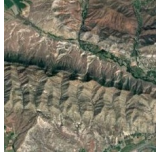
After fine-tuning the model, the results provided from BLEU calculations can be found on Table-I for baseline and fine-tuned model, and 5 samples from training set and predictions from each model can be found on Table-II.

The fine-tuned model achieves BLEU-4 of 0.318 compared to baseline 0.000, with consistent gains across n-gram orders (Table-I). The qualitative comparison (Table-II) shows two samples where fine-tuned output matches ground truth exactly, with the remaining three adopting RISC's characteristic phrasing patterns.

V. DISCUSSION

In this project, it was demonstrated that fine-tuning PaliGemma using QLoRA adapters can be successful for downstream tasks such as image captioning on satellite images. Even though the computation resources were not enough to finish 1 full epoch, the results shown on the Table-I and Table-II demonstrate high-level improvements. 2 of the 5 captions predicted by fine-tuned PaliGemma are identical to the ground truth, and the rest uses almost the same vocabulary as the ground truth captions. The baseline model's prediction results are generic, which is expected, because it is a general purpose pre-trained model. Fine-tuned captions also matches the length, declarativeness and phrasings of ground truth captions.

TABLE II
QUALITATIVE COMPARISON: BASELINE VS. QLoRA FINE-TUNED PALIGEMMA ON FIVE TEST IMAGES

Image	Ground Truth	Baseline	Fine-Tuned
	some planes are near several buildings in an airport.	Aerial view of the airport	many planes are parked in a row on the airport.
	A straight freeway goes through the lawn and some cars are driving on the freeway.	I-10 in Houston	The straight freeway goes through the lawn and some cars are driving on the freeway.
	The roundabout with some exits and entrances is in the commercial area and a lawn is in the middle of the roundabout.	Satellite view	The roundabout with some exits and entrances is surrounded by some buildings and a lawn is in the middle of the roundabout.
	The folded mountain consists of a number of ridges and valleys and is covered with some vegetation.	Satellite view	The folded mountain consists of some ridges and valleys and some vegetation is on the mountain.
	A basketball court next to a lake and some trees beside.	Satellite view	A basketball court next to a lake and some trees beside.

REFERENCES

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [2] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations*, 2021.